

CE 160D Final Report

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May, 2026

Abstract

This project, conducted in May 2026, presents a data-driven evaluation of aircraft Landing and Take-Off (LTO) emissions, with an intention to address their direct impact on the health of local fenceline communities. The study utilized a comprehensive dataset integrating 7.7 million domestic flights across 40 airports in 2025 from the Bureau of Transportation Statistics. This traffic data was merged with the ICAO Engine Emissions Databank for specific performance metrics, the FAA Civil Aircraft Registration Database for engine specifications, and the Open-Meteo Archive for historical hourly weather data. Emissions were calculated based on fuel consumption as a function of temperature, pressure, time-in-mode, and engine-specific thrust settings. Exploratory data analysis revealed that emissions are closely tied to aircraft type, airline fleet composition, and weather conditions, with higher temperatures and lower air density resulting in greater fuel burn. Using Principal Component Analysis (PCA) and K-Means Clustering, the research furthermore identified distinct airport emission profiles for the release of specific pollutants.

In conclusion, the analysis determined that NO_x is significantly more predictable than CO₂, with airline fleet composition and route distance serving as the strongest pre-flight predictors.

Introduction and Motivation

Greenhouse emissions from the aviation industry represent a large and challenging problem to solve in the modern world. Its function is directly linked to the success of our globalized and interconnected world, yet the reduction of its environmental impact is paramount to its continued existence and success. For communities nearby major airports, the local causation of emissions by air traffic have increasingly devastating impacts and are commonly associated with the landing and take off cycle, also known as the LTO cycle.

Emissions from the LTO cycle include those from aircraft approach and landing sequence, taxiing between terminals and runways, and include emissions generated while at the gate or other airport facilities.

The generation of harmful gasses at this regulated domain below 3000ft, become localized around an airport and its surrounding communities with tangible impacts on the people living around the communities. Specifically, emissions such as Nitrogen Oxides (NO_x) produced in the LTO cycle have shown to lead to increased risks in cardiovascular disease, respiratory distress, and pre-mature death.

In this study, we hope to identify various key characteristics, operational procedures, and airport traits that contribute to lower Landing and Take-Off (LTO) emissions at forty-one various US-airports. By evaluating the significant variability in emissions across different routes, carriers, and geographical locations, this research aims to break down the volume, chemical composition, and cycle stages of LTO phases. Ultimately, our goal is to translate these findings into a framework of actionable steps that can be implemented across the aviation industry. Our goal is to provide a roadmap for airports worldwide to mitigate their environmental footprint, protecting the health of

local communities and contributing to the global reduction of greenhouse gas emissions.

Data Sources and Data Processing

From the United States Department of Transportation’s Bureau of Transportation Statistics (BTS) we utilized their Airline Traffic Dataset from 2025. Within this data information on on-time aircraft performance was detailed and used to match with the following datasets:

We employed the use of a databank of aircraft engine emissions from the International Civil Aviation Organization (ICAO) hosted by the European Union Aviation Safety Agency (EASA). Engine manufacturers self-submit their data to this list during the certification of engines. Studied information includes turbojet and turbofan engines with a static thrust over 26.7 kilonewtons. Data such as engine types and emissions parameters were applied in the study.

To merge these two individual datasets further, the Aircraft Registration Database of the United States Federal Aviation Administration (FAA) was used to link tail numbers from the BTS Airline Traffic Dataset to aircraft type. Using aircraft type, we could then match engine configuration, number, type, efficiency, and emissions to the specific aircraft’s LTO emissions, which then could be linked to the corresponding airport.

Furthermore due to aircraft operations being highly affected by weather, and natural phenomena, we introduced a historical weather API from Open-Meteo which would assist in atmospheric temperature and pressure to support efficiency values in the FAA dataset.

As a result, we matched 1,532,364 different flights from 40 different airports in the United States, including Hawaii, Alaska, and Puerto Rico. The data set also includes 21 different carriers and 97 different types of aircraft.

Methodology

To calculate the emissions (E) for each phase of the LTO cycle (F) for each pollutant (P) and flight, we first need to find the fuel consumption (FC) for the aircraft during the phase so that we can multiply it with the appropriate emissions index (EI).

$$E_{F,P} = FC \cdot EI$$

To calculate fuel consumption, we use an equation found in Zou et al. (2025) and then multiply the fuel flow by a factor from Boeing’s Fuel Flow Method 2 that uses temperature and pressure to account for differences in fuel flow for different environmental factors, which in our case is differing elevations of airports as well as weather (Federal Aviation Administration 2016).

$$FC = \sum_{i \in F} \frac{\left(\frac{T}{518.67}\right)^{3.5}}{\left(\frac{\psi}{14.696}\right)} \cdot FF_i \cdot t_i \cdot n$$

Where FC is the fuel consumption, i is the flight phase, T is the Rankine temperature, ψ is the

pressure in psi, FF_i is the fuel flow for the flight phase, t_i is the time in the phase, and n is the number of engines. All of the aircraft in our data have 2 engines. The constants 518.67 Rankine and 14.696 psi represent the standard values at sea level.

To construct airport-level profiles, we aggregated flight-level data by computing mean LTO emissions, LTO phase breakdowns, operational features (taxi time, distance, air time), weather features, and fleet composition (Boeing and Airbus share). The features were then standardized using `StandardScaler` and reduced to two principal components via PCA. Finally, K-means clustering was applied separately for CO₂ and NO_x to identify groups of airports with similar emissions characteristics.

We also developed predictive models to evaluate whether pre-flight information alone can predict LTO emissions. Input features include route distance, origin/destination pair, airline, month, and geographic route type, with total CO₂ (ton) and total NO_x (kg) per flight as target variables. Four machine learning methods were evaluated: Linear Regression, Decision Tree, Gradient Boosting based models (XGBoost and LightGBM) using an 80/20 train-test split. Three model performance evaluation are used: R^2 , Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE). To assess feature importance, we used SHAP method within Gradient Boosting based models.

Results

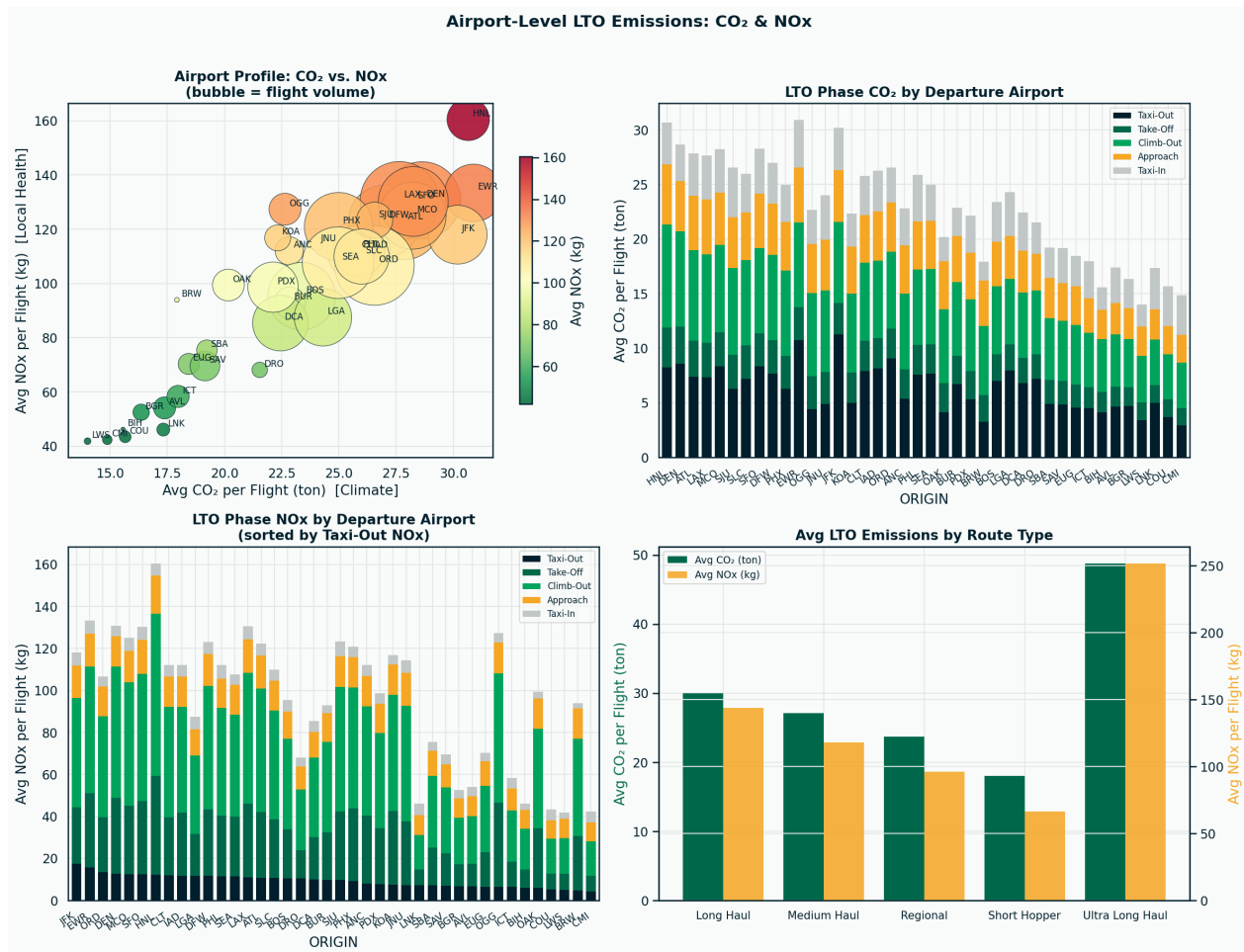


Figure 1: Airport-Level LTO Emissions

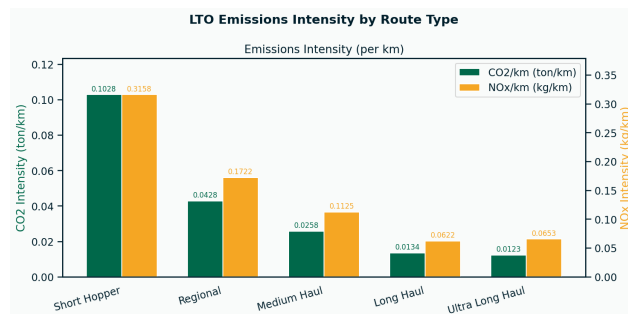


Figure 2: LTO Emissions per km

At the airport level, CO₂ and NO_x emissions exhibit a strong positive correlation across airports. Major hub airports (e.g., HNL, JFK, LAX) tend to have higher emissions per flight, while small regional airports show the opposite pattern. Notably, HNL shows the highest NO_x emissions

relative to CO₂, driven by its high volume of inter-island short-haul operations. In terms of LTO phase contributions, NO_x emissions are primarily driven by the Take-Off and Climb-Out phases, which require high engine thrust. CO₂, on the other hand, shows relatively higher contributions from ground operations such as Taxi-Out and Taxi-In. As expected, longer routes are associated with higher absolute emissions, reflecting greater total engine-on time. However, when considering emission intensity (ton/km), longer routes are more efficient, as fixed LTO-phase emissions are distributed over greater distances, resulting in lower emissions per kilometer.

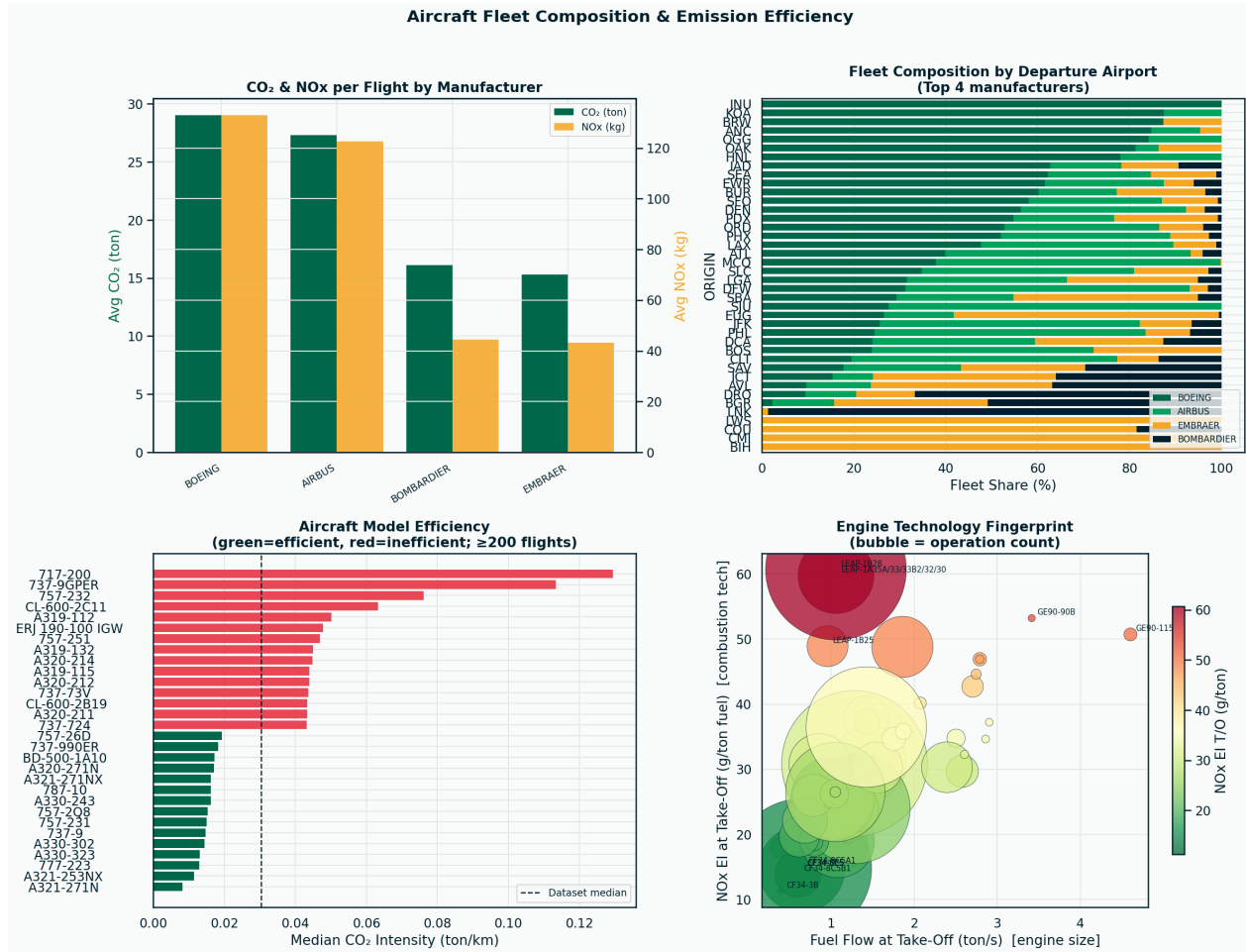


Figure 3: Aircraft Fleet composition and Emission Efficiency

At the aircraft level, Boeing and Airbus flights tend to show higher absolute emissions compared to regional jet manufacturers such as Bombardier and Embraer, as their fleets include not only short-range narrowbodies but also long-range widebody aircraft operated on longer routes. Fleet composition varies significantly by airport, some airports are dominated by specific manufacturers (e.g., JNU and KOA by Boeing, LNK by Bombardier). In terms of emissions efficiency, older aircraft types (e.g., Boeing 717-200) show significantly higher CO₂ intensity compared to newer narrowbodies (e.g., A321neo), reflecting improvements in engine and airframe technology. The Engine Technology Fingerprint reveals a trade-off between engine size and NO_x emissions, modern

high-efficiency engines such as the LEAP series show higher NO_x emission indices despite lower fuel flow, while older regional jet engines (e.g., CF34) exhibit lower NO_x EI with smaller engine size.

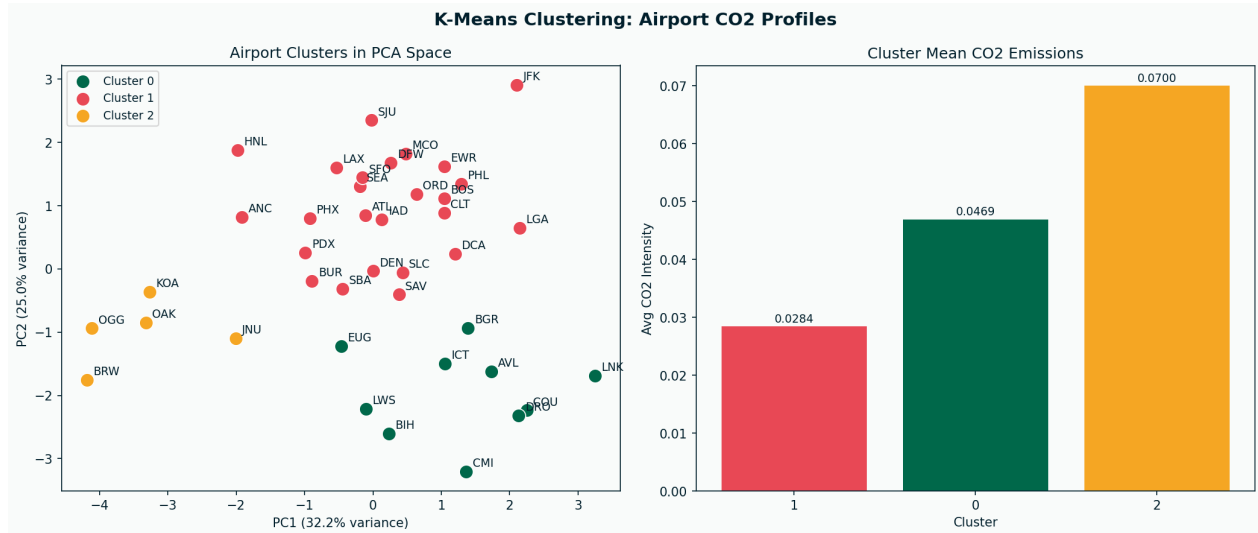


Figure 4: K-means clustering results of airport CO₂ emission profiles in PCA space (left) and mean CO₂ emissions per cluster (right).

As a result of K-means clustering on airport CO₂ emission profiles, three distinct clusters were identified. Cluster 0 (green) is primarily composed of small regional airports (e.g., AVL, BGR), characterized by the lowest average CO₂ emissions per flight (16.91 ton/flight), but moderate CO₂ intensity (0.0469 ton/km), as fixed LTO-phase emissions constitute a larger share of total emissions on short-haul routes (flights under 500 miles consist 86.2%). Cluster 1 (red) represents major hub airports with diverse networks and balanced fleets (e.g., JFK, SFO), exhibiting the highest absolute CO₂ emissions per flight (25.7 ton/flight) but the lowest CO₂ intensity (0.0284 ton/km), reflecting the efficiency gains of long-haul operations. Cluster 2 (orange) comprises Pacific Coast, Alaska, and Hawaii airports (e.g., BRW, OGG, OAK), unified by high Boeing fleet concentration (84.9%) primarily operated by Southwest, Hawaiian, and Alaska Airlines. Despite moderate absolute emissions (21.41 ton/flight), this cluster shows the highest CO₂ intensity (0.0706 ton/km). This elevated intensity is disproportionately driven by Hawaii airports (KOA, OGG), where Boeing 717-200 accounts for approximately 34% of flights on ultra-short inter-island routes, resulting in poor per-km efficiency despite moderate total emissions per flight.

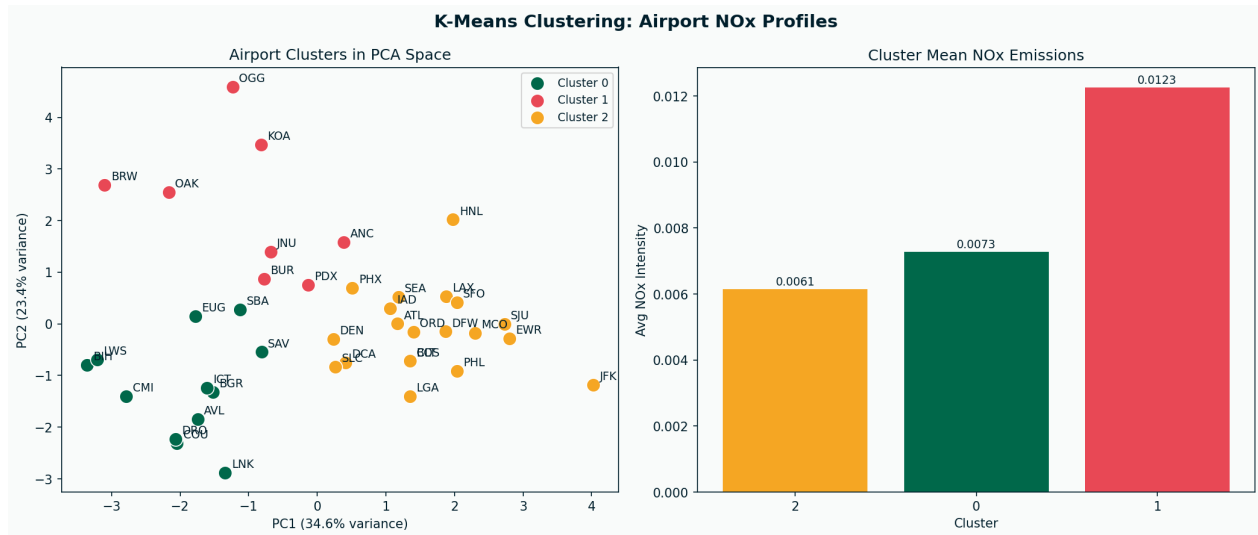


Figure 5: K-means clustering results of airport NOx emission profiles in PCA space (left) and mean NOx emissions per cluster (right).

As with NOx, Cluster 0 (green) is primarily composed of small regional airports (e.g., EUG, AVL), with the lowest NOx per flight (3.25 kg/flight) and moderate NOx intensity (0.0090 kg/km), as LTO-phase emissions constitute a larger share of total emissions on short-haul flights (81.2% under 500 miles). Cluster 2 (orange) represents major hub airports with broad networks and diverse fleets (e.g., DEN, LAX), exhibiting the highest NOx per flight (6.15 kg/flight) but the lowest NOx intensity (0.0078 kg/km), reflecting the efficiency of long-haul operations. Cluster 1 (red) consists of Alaskan (e.g., JNU, ANC), Hawaiian (e.g., KOA, OGG), and West Coast airports (e.g., OAK, PDX), unified by high Boeing fleet concentration (70.9%) primarily operated by Southwest, Alaska, and Hawaiian Airlines, with moderate NOx per flight (5.65 kg/flight) but the highest NOx intensity (0.0125 kg/km). This elevated intensity is disproportionately driven by Hawaii airports (KOA, OGG), where Boeing 717-200 accounts for approximately 34% of flights on ultra-short inter-island routes.

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FINAL SUMMARY						
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Model	CO2 R ²	CO2 RMSE	CO2 MAPE	NOx R ²	NOx MAPE	Time

Linear Regression	0.1489	9.38	26.98%	0.2307	42.50%	0s
Decision Tree	0.4838	7.31	17.42%	0.6256	19.77%	4s
XGBoost	0.5468	6.85	15.86%	0.6818	17.78%	8s
LightGBM	0.5631	6.72	15.31%	0.6983	16.70%	2m36s

The predictive models show that the performance gaps among the three tree-based models are relatively small (MAPE 17.42%, 15.86%, 15.31%) compared to Linear Regression (MAPE 26.98%), suggesting that non-linear relationships dominate LTO emission prediction. The best model, Light-

GBM, achieves R^2 of 0.56 for CO₂ and 0.70 for NO_x. Although NO_x shows higher R^2 , CO₂ achieves lower MAPE (15.31% vs. 16.70%), as MAPE is sensitive to the smaller absolute magnitude of NO_x values, where minor prediction errors result in disproportionately large percentage deviations. Therefore, neither target is strictly superior - CO₂ is more accurate at the individual flight level, while NO_x better captures overall variance patterns.

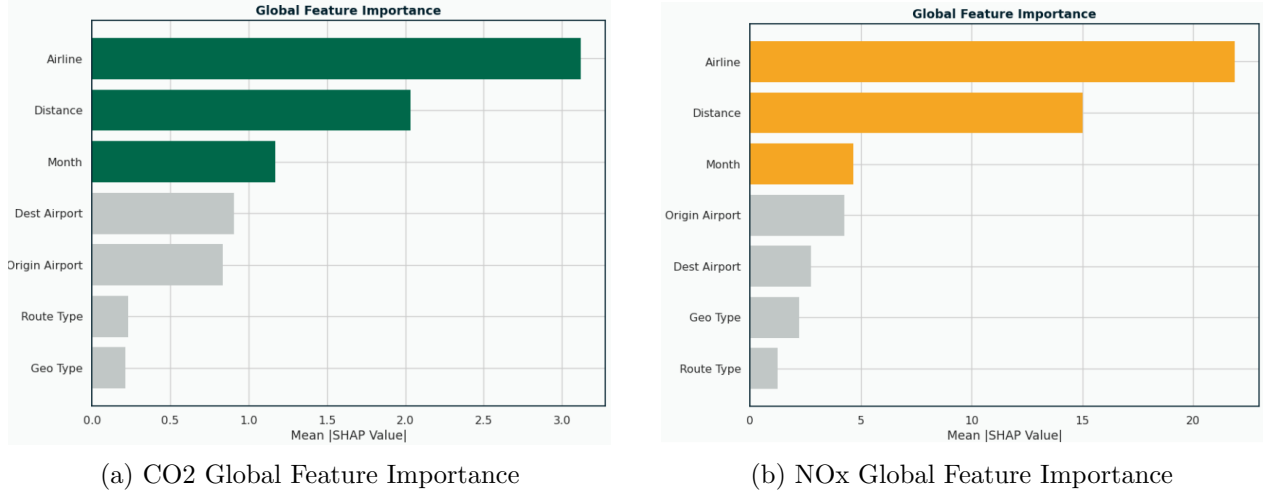


Figure 6: SHAP Global Feature Importance for CO₂ and NO_x prediction

SHAP feature importance from the LightGBM model reveals that Airline, Distance, and Month are the top three predictors for both CO₂ and NO_x emissions. The dominance of Airline as the most important feature reflects the fact that airlines tend to operate consistent fleet types and route networks, making carrier identity a strong proxy for aircraft type and mission profile. Distance is the second most important feature, directly influencing total fuel burn across LTO phases. Month captures seasonal variation in temperature and atmospheric pressure, both of which influence engine performance and fuel efficiency during LTO operations.

Discussion and Implications

Our research identified several physical and logistical factors that tie into baseline emissions at a variety of major hubs in the United States. At highly congested airports, such as JFK, aircraft often idle for 20 to 40 minutes after departing the gate and before take off during peak hours. This is supported by our data, where major hub airports show an average taxi-out time of 21.2 minutes, compared to 12.2 minutes for remote airports. During this time it is typical to see that aircraft wait for ATC clearance or taxi several miles between gates and active runways. As shown in our results, these factors both led to concentrated emissions of CO₂ during ground operations on the LTO cycle.

Further questions from this result challenge modern airline practices including those in place to minimize pollution. In similar studies, broad airport emissions from singular airlines are corroborated together, yet individualistic views stay unchecked. Investigations into whether airlines invest

directly into cleaner fleets instead of shuffling more high-emitting aircraft to smaller regional hubs to mask emissions mandates, are an interesting implication that should be pursued.

Future Improvement

Some gaps in our methodology include not calculating the emissions of aircraft while at the gate, which is a part of the LTO cycle we have omitted, and accounting for more variables such as visibility, wind speed, fuel type, peak travel periods, and more.

We can also add aircraft seat capacity data to normalize emissions on a per-seat or per-passenger level, enabling more deeper comparison across aircraft types and routes types. For example, a large wide-body aircraft may be less efficient in absolute volume, but more efficient per passenger compared to small narrow-body aircraft.

References

- Agency for Toxic Substances and Disease Registry (2024). *ToxFAQs for Nitrogen Oxides*. CDC. URL: <https://www.cdc.gov/tsp/ToxFAQs/ToxFAQsDetails.aspx?faqid=396&toxid=69> (visited on 05/08/2026).
- Bureau of Transportation Statistics (2025). *2025 Statistics*. URL: https://www.transtats.bts.gov/DL_SelectFields.aspx?gnoyr_VQ=FGK&QO_fu146_anzr=b0-gvzr (visited on 05/08/2026).
- Federal Aviation Administration (2016). *Aviation Environmental Design Tool (AEDT) Technical Manual, Version 2b, Service Pack 2*. Technical Manual DOT-VNTSC-FAA-16-04. Washington, D.C.: U.S. Department of Transportation. URL: <https://rosap.ntl.bts.gov/view/dot/36039>.
- (2026). *Civil Aviation Registry, Releasable Aircraft Database*. FAA. URL: https://www.faa.gov/licenses_certificates/aircraft_certification/aircraft_registry/releasable_aircraft_download (visited on 05/08/2026).
- Open-Meteo (2026). *Historical Weather API*. URL: <https://open-meteo.com> (visited on 05/08/2026).
- Zou, Runyuan et al. (2025). “A pathway to sustainable aviation: Modeling aircraft takeoff mass for precise fuel consumption and aircraft emission calculations”. In: *Energy* 319, p. 135074. DOI: [10.1016/j.energy.2025.135074](https://doi.org/10.1016/j.energy.2025.135074).